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RV COLLEGE OF ENGINEERING®
Autonomous Institution affiliated to VTU
V Semester B. E. April 2024 Examinations
DEPARTMENT OF INDUSTRIAL ENGINEERING AND MANAGEMENT
Model Question Paper
COURSE: DIGITAL METROLOGY
(2021 SCHEME)

Time: 03 Hours**Maximum Marks: 100****Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8 and 9 and 10.

PART – A

1			M	BTL	CO
	1.1	Differentiate between static and dynamic errors in the context of measurements.	02	1	1
	1.2	Give two examples each for resistive and capacitive transducers	02	2	2
	1.3	Differentiate between A/D and D/A converters in the context of a data acquisition system.	02	2	2
	1.4	Give two applications of velocity and motion	02	2	2
	1.5	Differentiate and Hole and shaft basis system.	02	2	3
	1.6	List the important types of fits.	02	1	3
	1.7	Give the working principle of optical flats and Tool Makers microscope.	02	2	2
	1.8	Define Best size wire.	02	1	2
	1.9	List any four applications of CMM	02	2	4
	1.10	Differentiate between bridge and gantry type CMM.	02	2	4

PART – B

UNIT – I					
2	a	Discuss the components and functioning of the generalized measurement system, providing justification through an illustrative example, clear sketches, and block diagrams	08	3	1
	b	Describe briefly the working principle of a Capacitive Transducer and also explain the measurement of linear displacement in a using the change in the area of Plates.	08	2	1

UNIT - II					
3	a	Discuss the fundamental concept of sensors and their importance in metrology. Include examples to elucidate various types of sensors employed in measurement procedures .	08	3	2

	b	Explain the operational principles of pyroelectric sensors and their application in metrology for the detection of infrared radiation and temperature variations.	08	2	2
OR					
4	a	Classify sensors based on their functionality into displacement, position, and proximity sensors. Discuss any one type, providing examples and applications in metrology.	08	3	2
	b	Explain how piezoelectric sensors are employed for fluid pressure measurements in metrology. Provide examples of their usage and advantages in pressure sensing applications.	08	3	2

UNIT - III					
5	a	Calculate the limits of tolerance and allowance for a 25mm shaft and hole pair designated by H ₈ d ₉ and sketch the fit. Given: 25mm lies between 18 and 30, Fundamental deviation of 'H' hole=0, fundamental deviation of 'd' shaft=-16D 0.44 ,IT8=25i,IT9=40i	08	4	4
	b	Discuss the concepts of (i) Interchangeability and (ii) Selective Assembly, highlighting their significance in engineering and manufacturing processes.	08	2	3
OR					
6	a	Define the concept of fit and provide a concise overview of the different types of fits, supplemented with illustrative examples.	08	2	3
	b	Calculate the relevant dimensions of 35H ₇ f ₈ fit, dimensions 35 falls in the steps of 30-50mm. Fundamental deviation for f shaft is -5.5D 0.41 ,IT7=16i,IT8=25i	08	4	4

UNIT - IV					
7	a	Illustrate the terminologies utilized to denote surface roughness through a clear diagram, elucidating each term's significance.	08	3	2
	b	Illustrate the operational mechanism of a floating carriage micrometer using a clear diagram and provide a concise explanation of its functionality.	08	2	2
OR					
8	a	Enumerate and describe five frequently employed techniques for measuring surface roughness in engineering contexts.	08	3	2
	b	Illustrate the operational principles of a tool maker's microscope through a clear diagram, providing a brief overview of its functionality.	08	2	2

UNIT - V					
9	a	Illustrate and describe two variations of Coordinate Measuring Machines (CMMs) using sketches to elucidate their distinguishing features and operational methodologies.	08	2	4
	b	Describe briefly the working of a Michelson Interferometer and list out its applications.	08	2	3
OR					
10	a	Discuss the significance and applications of Coordinate Measuring Machines (CMMs) in various industries.	08	3	4
	b	Explain the fundamental principle underlying LASER technology. Enumerate the diverse applications of LASER in metrology and elaborate on two specific applications in depth.	08	2	3



.....Q.P.Code (For office use only)

Course Code: 21IM52	Course Title: Digital Metrology
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PART-A

Q.No	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10
B T	1	2	2	2	2	1	2	1	2	2
COs	1	2	2	2	3	3	2	2	4	4
Q No	1.11	1.12	1.13	1.14	1.15	1.16	1.17	1.18	1.19	1.20
B T										
COs										

PART-B

Question No		B T Levels	Cos addressed		Question No		BT Levels	Cos addressed
2	a	3	1		3	a	3	2
	b	2	1			b	2	2
	c					c		
	d					d		
4	a	3	2		5	a	4	4
	b	3	2			b	2	3
	c					c		
	d					d		
6	a	2	3		7	a	3	2
	b	4	4			b	2	2
	c					c		
	d					d		
8	a	3	2		9	a	2	4
	b	2	2			b	2	3
	c					c		
	d					d		
10	a	3	4		11	a		
	b	2	3			b		
	c					c		
	d					d		

Signature of Scrutinizer:

Signature of Chairperson:

Name:

Name: